



UNIVERSITY OF THE PUNJAB

Seventh Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistical Inference-I (Theory)
Course Code: STAT-401

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Attempt all questions.

Q. No. 2. Write a short note on the following:

(5 each)

- Relative Efficiency
- Risk Function
- Drawbacks of Factorization Theorem
- Exponential Family of Distributions

Q. No. 3. Let X has mean p and variance pq and $S_n = X_1 + X_2 + \dots + X_n \sim B(n, p)$.

(08)

- Prove that $\frac{S_n}{n}$ is an unbiased estimator of p . show that $E[1 - \frac{S_n}{n}]$ is equal to $1 - p = q$. But $\frac{S_n}{n}(1 - \frac{S_n}{n})$ is not an unbiased estimator of pq . Find the amount of bias?
- Prove that $\frac{S_n}{n}(1 - \frac{S_n}{n})$ is asymptotically unbiased estimator of pq .
- Prove that $\frac{S_n}{n}$ is consistent in mean square.
- Prove that $\frac{S_n}{n}(1 - \frac{S_n}{n})$ is consistent estimator of pq .

Q. No. 4. Given that the pdf of uniform distribution as

(07)

$$f(x; \theta) = \frac{1}{\theta} \quad 0 < x < \theta, \quad -\infty < \theta < +\infty$$

Show that $Y_n = \max(X_i)$ is a sufficient statistic.

Q. No. 5. Given that $f(x; \theta) = \frac{1}{\theta} \quad 0 < x < \theta, \theta > 0$
= 0 elsewhere.

(07)

Prove that the C.R.L.B is less efficient than the estimator $(\frac{n+1}{n})y_n$, where y_n is the largest value of a random sample of size n and y_n is not unbiased. Suppose its variance is

$$E(y_n^2) - [E(y_n)]^2.$$

Q. No. 6. State and prove Rao Blackwell theorem.

(08)

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Attempt this Paper on Separate Answer Sheet provided.

Attempt all questions.

- Q. No. 2. Write a short note on the following: (5 each)
- Weak and strong consistency
 - Regularity conditions
 - Loss Function
 - Bayes Estimator

- Q. No. 3. Let $X_1, X_2, \dots, X_6$ is an identically independently distributed random sample from normal distribution with mean μ and variance σ^2 . Determine 'c' such that $c[(X_1 - X_2)^2 + (X_3 - X_4)^2 + (X_5 - X_6)^2]$ is an unbiased estimator of σ^2 . (07)

- Q. No. 4. Let $y_1 < y_2 < \dots < y_n$ denote the order statistic of a random sample $X_1, X_2, \dots, X_n$ from the distribution that has p.d.f (07)

$$f(X; \theta) = e^{-(x-\theta)} \quad \theta < x < \infty, \quad -\infty < \theta < +\infty$$

Assume $n = 3$ and show that $\max(X_i)$ is not a sufficient statistic for θ .

- Q. No. 5. If 't' is a Minimum Variance Unbiased Estimator of $\tau(\theta)$ with variance v_1 and T_1 & T_2 are any two unbiased estimators of $\tau(\theta)$ with finite variances. (08)
- If either T_1 or T_2 is minimum variance estimator. Then show that correlation coefficient between T_1 and T_2 is $E^{1/2}$ (where 'E' is the relative efficiency of T_1 and T_2).

- Q. No. 6. Show that if minimum variance estimator exists, it would essentially be unique (08)
- irrespective of whether any bound is attained.

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UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program : Seventh Semester – 2020

Paper: Statistical Inference-I (Theory)
Course Code: STAT-401

Part – II

Time: 2 Hrs. 45 Mins.

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following:

- i) What is the problem of estimation? How it can be dealt? (05)
- ii) Differentiate between i) Asymptotic Unbiasedness and Consistency ii) Efficiency and Mean Square Consistency. (06)
- iii) Compare the Best asymptotically Normal Estimator, Minimum Variance Bound Estimator and Efficient Estimator (06)
- iv) State Lehmann-Sheffe's theorem. (03)

Answers the following questions.

(3x10=30)

- Q.3. Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from uniform distribution $U(0, \theta+1)$ and let $Y_1 < Y_2 < Y_3$ be the order statistic of order 1. Check Y_1 for possible unbiasedness. (10)
- Q.4. Prove that minimum variance bound estimator is Unique. (10)
- Q.5. State and prove Rao-Blackwell theorem. (10)

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UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Seventh Semester – Fall 2021

Roll No.

Paper: Statistical Inference-I (Theory)

Course Code: STAT-401

Time: 3 Hrs. Marks: 60

Q.1. Answer the following short questions:

- i) Write a note on the comparison of Asymptotic Unbiasedness and Consistency. (05)
- ii) Write the procedures for finding whether any estimator is sufficient or not, what are (if any) the objections on them? (4+4)
- iii) How would you explain the joint sufficiency? (02)
- iv) Differentiate between location invariant and scale invariant estimators. (03)
- v) What do you mean by completeness? explain. (05)
- vi) Write and explain different forms of efficiency. (07)

Q.2. Answer the following questions.

- i) Let $f(x; \theta, p) = \frac{1}{\Gamma(p)\theta^p} x^{p-1} e^{-x/\theta}$, $0 < x \leq \infty$, $p > 0$. The M.V.B estimator of θ for known p is $\bar{x}/p$ with variance $\theta^2/(np)$, find the M.V.B estimator of 'p' and its variance. (10)
- ii) Prove that sufficient statistic is unique. (08)
- iii) Let $x_1, x_2, x_3, \dots, x_n$ be a random sample, n being fixed, from a uniform distribution over the range $(0, a)$. Check $\hat{M} = \max(x_1, x_2, \dots, x_n)$ and $\hat{Q} = \min(x_1, x_2, \dots, x_n)$ for possible unbiasedness and consistency. (12)
Where $\hat{M}$ and $\hat{Q}$ are order statistics



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program ; Seventh Semester – 2021

paper: Statistical Inference-I (Theory)
Course Code: STAT-401

Part – II

Roll No.

Time: 2 Hrs. 45 Min. Marks: 80

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following:

- i) What are the problems of estimation? (05)
- ii) Differentiate between (4+4)
 - a) Asymptotic Unbiasedness and Consistency
 - b) Efficiency and Mean Square Consistency.
- iii) How would you explain the joint sufficiency? (02)
- iv) Differentiate between location invariant and scale invariant estimators. (02)
- v) What do you mean by completeness? Explain. (03)

Answers the following questions.

(3x10=30)

Q.3. Let $Y_1 < Y_2 < Y_3$ be the order statistics of random sample of size 3 from uniform (10)
distribution over the interval $[0, \theta]$. Check the unbiasedness of these three
statistics, how they can be made to be unbiased if some or all of these
statistics are biased.

Q.4. State and prove Uniqueness theorem for efficiency. (2+8)

Q.5. Let $t = x_1 + x_2 + x_3$, $k = x_1 + 2x_2 + x_3$ be two statistics where x_i follow Bernoulli (10)
distribution, check whether or not the 't' and 'k' are sufficient by using
conditional probability approach.

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UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2024

Paper: Statistical Inference-I (Theory)

Course Code: STAT-401

Roll No.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- I. ✓ What do you know about the efficient estimator?
- II. ✓ How do you explain the joint sufficiency?
- III. ✓ Define the best asymptotically normal estimators (BAN)
- IV. ✓ State draw backs of factorization theorem
- V. Define complete statistics and complete parameter families
- VI. State Rao-Blackwell Theorem

Q.2. Answer the following questions.

(3x10=30)

- I. State and prove Cramer- Rao inequality for minimum variance of an estimator.
- II. Let $t = x_1 + x_2 + x_3$, $k = x_1 + 2x_2 + x_3$, be two statistics where x_i follow Bernoulli distribution, check whether or not "t" and "k" are sufficient by using conditional probability approach
- III. For the two parameters of Gamma distribution with p.d.f

$$F(x) = \frac{x^{\alpha-1}}{\alpha \beta^{\alpha}} e^{-\frac{x}{\beta}} \quad 0 < X < \alpha$$

IF X is known, then show that β^x is equal to $\frac{x}{\alpha}$ is M.V.B.E parameter of unknown parameter β



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2022

Paper: Statistical Inference-I (Theory)

Course Code: STAT-401

Roll No. 032456

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- What are the problems of estimation? Explain each.
- What is the importance of sufficiency and joint sufficiency as a property of good estimator?
- The Cramer-Rao inequality is basis of efficiency; discuss briefly?
- Compare best asymptotically normal estimator with efficient estimator.
- Differentiate between location and scale invariant estimators.
- How the property of unbiasedness and efficiency are related? Which one is more needed?

Q.2. Answer the following questions.

(3x10=30)

- Let $t = x_1 + x_2$ $k = x_1 + 2x_2$ be two statistics where x_i follow Poisson distribution, check 't' and 'k' for possible sufficiency.
- With reference to Cramer-Rao inequality and with usual notations, prove that

a) $E[(\partial \ln L)/(\partial \theta)] = 0$, b) $E[(\partial \ln L)/\partial \theta]^2 = -E[(\partial^2 \ln L)/\partial \theta^2]$.

- State and prove Lehmann-Sheffe theorem.

UNIVERSITY OF THE PUNJAB, LAHORE.

EXAMINATION: B.S 4 YEARS PROGRAMME

PAPER: Statistical Inference-I (LAB) Group-II

Course code: STAT-402

Time: 03 Hrs

Max Marks: 100

NOTE: Attempt ALL questions.

Q.No.1:

(40)

Draw 40 random samples of size 3 from a Uniform Distribution i.e. $X \sim U(2,5)$.

Find sample mean and sample median from each sample and:

- Check that these are unbiased estimators;
- By finding variance of these statistics, which is more efficient?

Q.No.2:

(40)

Let X_1, X_2, X_3 be a random sample of size 3 from Bernoulli distribution. Consider following statistics:

$$T = X_1 + X_2 + X_3$$

$$Z = (X_1 + 2X_2 + 3X_3)/6$$

Using Conditional Probability Criteria of Sufficiency, Check T and Z for Sufficiency.

Q.No.3: Practical Note book

(10)

Q.No.4: Viva Voce.

(10)



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- Define Estimator, Estimate and Estimation with examples.
- Describe the types of Consistency.
- How Risk and Loss functions are related?
- Define exponential family of Distribution.
- Describe the role of Mean-squared error in Estimation Theory.
- What is meant by Sufficiency? Explain the Fisher-Neyman factorization criterion for sufficiency.

Q.2. Answer the following questions.

(3x10=30)

1) Let $Y_1 < Y_2 < Y_3$ be the order statistics of random sample of size 3 from uniform distribution over the interval $[0, \theta]$. Show that $4Y_1$, $2Y_2$ and $\left(\frac{4}{3}\right)Y_3$ are all unbiased Estimators for theta with respect to variances $\frac{3\theta^2}{5}$, $\frac{\theta^2}{5}$ and $\frac{\theta^2}{15}$.

2) State and prove Cramer-Rao inequality for minimum variance of an estimator.

3) Let $Y_1 < Y_2 < \dots < Y_n$ is an order statistics. Let the distribution be $f(x; \theta) = e^{-(x-\theta)}$.

$$\theta < x < \infty$$

$$-\infty < \theta < \infty$$

Show that Y_1 is a sufficient statistic and Y_n is not a sufficient statistic.